Public health-specific personal disaster preparedness training: An academic-practice collaboration

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ABSTRACT

Objectives: *To measure the following three relevant outcomes of a personal preparedness curriculum for public health workers: 1) the extent of change (increase) in knowledge about personal preparedness activities and knowledge about tools for conducting personal preparedness activities; 2) the extent of change (increase) in preparedness activities performed post-training and/or confidence in conducting these tasks; and 3) an understanding of how to improve levels of personal preparedness using the Extended Parallel Process Model (EPPM) framework.*

Design: *Cross-sectional preinterventional and postinterventional survey using a convenience sample.*

Setting: *During 2010, three face-to-face workshops were conducted in three locations in West Virginia.*

Participants: *One hundred thirty-one participants (baseline survey); 69 participants (1-year resurvey)—representing West Virginia local health department (LHD) and State Health Department employees.*

Interventions: *A 3-hour interactive, public health-specific, face-to-face workshop on personal disaster preparedness.*

Main outcome measure(s): *Change in 1) knowledge about, and tools for, personal preparedness activities; 2) preparedness activities performed post-training and/or confidence in conducting these activities; and 3) the relationship of EPPM categories to personal preparedness activities.*

Results: *One year postworkshop, 77 percent of respondents reported having personal emergency kits (40 percent at baseline) and 67 percent reported having preparedness plans (38 percent at baseline) suggesting some participants assembled supply kits and plans postworkshop. Within the context of EPPM, respondents in high-threat categories agreed more often than respondents in low-threat categories that severe personal impacts were likely to result from a moderate flood. Compared to respondents categorized as low efficacy, respondents in high-efficacy categories perceived confidence in their knowledge and an impact of their response on their job success at higher rates.*

Conclusions: *Personal disaster preparedness trainings for the LHD workforce can yield gains in relevant preparedness behaviors and attitudes but may require longitudinal reinforcement. The EPPM can offer a useful threat and efficacy-based lens to understand relevant perceptions surrounding personal disaster preparedness behaviors among LHD employees.*

Key words: disaster, personal, preparedness, public health, Extended Parallel Process Model

INTRODUCTION

In the face of natural disasters worldwide, large-scale terrorist events, and emerging pandemic

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threats, government and response agencies have emphasized the need for personal emergency preparedness. For example, the Federal Emergency Management Agency (FEMA) launched its national *Ready* campaign in 2003. The campaign aims to increase public awareness and participation in disaster preparedness throughout the United States. *Ready* promotes three preparedness activities about different types of emergencies and their appropriate responses: "get a kit, make a plan and be informed."[1]

Despite government efforts, numerous studies and polls suggest low personal preparedness levels in the community.[2-7] A 2007 survey by the National Center for Disaster Preparedness suggests 47 percent of American adults believe they will personally experience a major disaster in the next 5 years, yet only 34 percent have engaged in personal preparedness activities.[6] A 2009 American Red Cross influenza A (H1N1-09) poll suggests only 46 percent of Americans have enough food, water, and medicine to allow them to remain home for up to 2 weeks if ill.[7]

Local health departments (LHDs) are at the heart of the public health emergency preparedness system (PHEPS).[8] In this context, personal preparedness comprises the "readiness" dimension of a "ready, willing, and able" workforce.[9] Several studies have assessed LHD workers' willingness to respond (WTR) to a disaster.[9-14] Research suggests that WTR is scenario-dependent, distinct from "ability" (ie, knowledge and skills), and is modified by a variety of risk perception-based cognitive and affective factors and attitudes/beliefs.[11,13,15,16] Importantly, among these influential modifiers of WTR, several can be directly linked to personal preparedness, including concern for the safety of dependent family members, apprehension about arriving to work safely, lack of self-efficacy, and a lack of understanding of public health emergency threats that may occur in their local region.[11,13,15,16] Gaps in personal preparedness planning among LHD workers could thus pose a significant barrier to public health emergency surge capacity in the face of an ever-widening array of threats.

Recently, fiscal austerity considerations and staffing constraints have starkly impacted this vital frontline response cohort.[17] A 2010 survey of US LHDs found that these agencies lost approximately 15 percent of their workforce due to budget cutbacks in the preceding year alone.[18] Additionally, nearly half the agencies surveyed anticipated they would be affected by further budget cuts. These intensifying constraints on human and fiscal resources imperil local public health surge capacity to respond to disasters—and by close extension, health security. Given the need to meet all-hazards response expectations amidst budget-related staffing shortfalls, the level of personal disaster preparedness of the LHD workforce has sizeable implications for the communities they serve.

An expanding literature points to Witte's threat and efficacy-based Extended Parallel Process Model (EPPM)[19] as a framework for understanding WTR among a variety of actors within the PHEPS.[11,13,14] These studies support the EPPM's central premise that people are most apt to constructively address a given hazard (in this case, willingness to come to work and respond to an incident) if they perceive its attendant threat as legitimate and motivating, and view their own response activities as meaningful and efficacious.

In late 2007, the Johns Hopkins Center for Public Health Preparedness (JH-CPHP) collaborated with the West Virginia Bureau for Public Health-Center for Threat Preparedness (WVBPH-CTP) to assess the willingness and personal preparedness of LHD employees to respond to public health emergencies. Thirty-four local (six regional) West Virginia LHDs participated in the survey using the *Johns Hopkins~Public Health Infrastructure Response Survey Tool* (*JH~PHIRST*).[11] The survey measured respondents' perceptions including awareness, skills, and confidence in performing assigned tasks; WTR to a public health threat; beliefs about their and their family's safety when responding to a public health emergency; and perceived efficacy and importance of their roles in combating a pandemic threat.

Survey results pointed to a gap in West Virginia's public health responders' personal preparedness. Eighty-five percent of respondents indicated they had a role in response to a public health emergency and 68 percent indicated they had family members dependent on them, yet only 37 percent reported having a family emergency kit.[11] This gap was noted by senior staff at the West Virginia Department of Health and Human Resources (WVDHHR) and raised concerns about its potential impact on the willingness and readiness of employees to report to work during a regional incident.

In this article, we describe the novel application of results from the 2007 EPPM-centered survey to develop an evidence-based approach for addressing gaps in personal disaster preparedness planning among West Virginia public health workers, and we characterize 1-year postcurriculum impacts on relevant behaviors and attitudes. The aims of the study were to measure three relevant outcomes of a personal preparedness curriculum for public health workers: 1) the extent of change (increase) in knowledge about personal preparedness activities and knowledge about tools for conducting personal preparedness activities; 2) the extent of change (increase) in preparedness activities performed post-training and/or confidence in conducting these tasks; and 3) an understanding of how to improve levels of personal preparedness using the EPPM framework.

METHODS

JH-CPHP-WVBPH-CTP collaboration

In February 2010, an employee survey was conducted across all 49 West Virginia LHD administrators to assess interest levels in providing personal preparedness training to LHD workers. Forty-six LHDs responded to the survey. Eighty-four percent of the administrators expressed interest in the training and 87 percent reported they would also support sharing materials with community public health responders. Senior management contacted JH-CPHP, proposing a collaboration to create a personal preparedness workshop for their employees and for accomplishing key practical preparedness steps during the training. The developed curriculum was entitled "Think Inside the Box."

Workshop development

JH-CPHP experts adapted existing FEMA *Ready* content into an interactive, public health-specific, face-to-face module. Content developers comprised two subject-matter experts with a combined 15 years experience in the field of public health preparedness and training, and a third member with expertise in educational design who independently reviewed the curriculum.

JH-CPHP's experts tailored existing FEMA materials into a public health-specific context and developed a scenario-based exercise—"West Virginia Severe Flooding Tabletop." This exercise systematically portrayed the individual and professional importance of personal preparedness within a public health context. Specifically, "Individual" here refers to the importance of preparedness within the context of an individual's personal life—family, children, and the physical household. "Professional" refers to the way in which preparedness affects ability and motivation to perform in a professional capacity—worker's availability, state of mind, and WTR to an incident created by anxiety for family members and other household concerns. Flooding was chosen as it has been identified by FEMA as one of West Virginia's most common weather-related risks.[15] The hypothetical scenario depicted severe flooding along the Ohio River from West Virginia to Illinois, periodically jumping forward in time to cover a 4-day period. Within the exercise, public health-related "injects" for discussion were based on several real incidents that previously occurred in West Virginia. This exercise encouraged participants to discuss personal and professional considerations of a flood scenario and to talk through plans and problems.

Workshop delivery

Between May and June 2010, three identical workshops were conducted with participating LHDs in three locations in West Virginia. Attendance at the workshops was voluntary and was encouraged by LHD management. One of the abovementioned JH-CPHP experts assumed responsibility as course trainer. The resulting workshop consisted of a 2-hour slide presentation adapted from FEMA *Ready* materials and a 1-hour "West Virginia Severe Flooding Tabletop."[1,20] The workshop included discussions and hands-on activities to aid participants in starting an emergency preparedness kit. Participants were also given templates, instructions, and time to start working on a family communication plan.

The workshop concluded with a trivia game where participants were asked a series of public health preparedness questions. "Winning" teams received FEMA-recommended household preparedness supplies such as a flashlight or a whistle. At the end of the game, all participants received several basic supplies to promote the assembly of a preparedness kit.

Data collection and analysis

Research ethics approval for the survey and its administration was received from The Johns Hopkins Bloomberg School of Public Health (JHSPH Institutional Review Board) with a waiver of written consent. Data collected during the workshops consisted of a paper-based, four-part, voluntary, anonymous questionnaire administered pretraining and immediately post-training (henceforth called baseline and post-test 1, respectively). The four parts included 1) standard demographic information, 2) eight questions measuring aspects and level of personal preparedness and confidence, 3) eight questions measuring knowledge of emergency preparedness, addressing core concepts of FEMA preparedness materials and important information addressed during training, and 4) eight attitude/belief statements regarding a moderate flood scenario. The questionnaire was adapted from the *JH~PHIRST* survey used with other LHD cohorts.[11,21,22]

Knowledge questions had four possible responses, only one of which was correct. An "adequate knowledge" designation was assigned to those with at least six correct responses. Demographic characteristics included gender, age, highest education level, professional classification, years in the current organization, and years in the current profession.

Attitude/belief statements addressed participants' perceptions of the likelihood of a moderate flood in their region and its potential consequences, perceptions of their confidence in performing protective actions, their perceived efficacy, and importance of their role in combating this threat. Responses to attitude/belief statements were based on a nine-point Likert scale of agreement with the statement. Prior to analyses, responses were dichotomized into categories of agreement versus disagreement and neutrality. Missing and "don't know" responses were excluded from analyses. Fisher's exact tests were used to determine which factors, demographic or knowledge-level, were highly associated with a positive (agreement) response for each attitude/belief statement.

One of four EPPM categories was assigned to each respondent based on each survey separately: low threat and low efficacy (LT/LE); low threat and high efficacy (LT/HE); high threat and low efficacy (HT/LE); and, high threat and high efficacy (HT/HE). "Threat" was determined as the product of a respondent's perceived likelihood of the occurrence of a moderate flood (item 1, Table 4) and the perceived severity of the event (item 2). "Efficacy" was calculated as the product of the respondent's perceived ability to perform their duties (item 5) and the perceived importance of their role in combating this threat (item 6). Low and high categories of threat and efficacy were determined by the median values of these two products. Percent agreement on attitude/belief statements was calculated within the four EPPM categories. Marginal comparisons were also made between the threat categories (HT vs LT) and between the efficacy categories (HE vs LE).

Tracking pretraining and post-training responses by subject was not available because subject identifiers were not used on the questionnaires. Analyses were thus performed separately for responses from baseline and post-training questionnaires. All analyses were performed using STATA version 11.1 (STATA Statistical Software; College Station, TX; Stata Corp LP, 2010).

Course evaluation

At the end of each workshop, participants provided feedback through an evaluation form containing nine statements relating to course content and delivery and a section for additional comments. Participants indicated their level of agreement with each statement based on a four-point Likert scale.

One-year follow-up

From April 15 to June 16, 2011, 1 year after the delivery of the three workshops, a link to Survey Monkey (*SurveyMonkey.com*, Portland, OR) was distributed by WVBPH-CTP collaborators to all original workshop participants via e-mail (henceforth called post-test 2). Respondents were asked to anonymously reply to the same questions included in the earlier surveys, and responses were analyzed similarly.

RESULTS

One-hundred thirty-one workshop participants, representing 35 LHDs and the West Virginia State Health Department (WVSHD), completed the baseline questionnaire and 129 of these participants (from 35 LHDs and WVSHD) completed the post-test 1 questionnaire. Sixty-nine of the participants, representing 26 LHDs and WVSHD, completed the 1-year follow-up questionnaire (post-test 2).

Demographic characteristics

Table 1 displays respondent demographic characteristics for each survey. Most respondents were female (64 percent at baseline). Age groups were well-represented. Nearly half of the respondents had ≥10 years experience in their current profession (47 percent) and 73 percent held at least a Bachelor's degree as their highest degree. The demographic composition at post-tests 1 and 2 was similar.

Level of personal preparedness

Over time, the percentage of respondents who reported having family emergency kits increased (41 percent baseline, 69 percent post-test 1, 77 percent post-test 2; Table 2) as did the percentage of respondents who reported having a family emergency plan (38 percent, 53 percent, 67 percent, respectively). The percentage of respondents answering "yes" to "I feel confident" questions (items 2-3 and 5-8, Table 2) ranged from 73 percent to 92 percent at baseline, 98 percent to 99 percent at post-test 1, and 97 percent to 100 percent at post-test 2. The levels of missing responses across all items in Table 2 ranged from 1 percent to 5 percent (data not shown).

Personal preparedness knowledge

At baseline, 81 percent of respondents met criteria for adequate knowledge regarding emergency preparedness (ie, ≥75 percent correct answers in Table 3). At post-test 1, this percent increased to 97 percent and decreased to 83 percent at post-test 2. A similar pattern of an increase followed by a decrease in correct answers over time was observed on many individual questions (Table 3). On the other items, respondents scored consistently high. For example, when asked about necessities for an effective household response to a winter storm, more than 98 percent had the correct answer on all three surveys.

Attitude/belief statements

Percent agreement with the eight attitude/belief statements was high across all survey administrations (Table 4). The most marked change between surveys was in the perceived personal consequences resulting from a moderate flood: 51 percent at baseline, 81 percent at post-test 1, and 50 percent at post-test 2. Sizeable changes were also observed over time in the percent perceiving public health severity of the event: 91 percent at baseline, 96 percent at post-test 1, and 58 percent at post-test 2. The percent perceiving a high likelihood of flood in their region ranged from 89 percent to 96 percent. Most respondents felt confident in their knowledge and ability to perform their duties and were confident that the successful performance of protective actions would have a positive impact on job success and overall success (percent agreement ranged from 85 percent to 96 percent). Respondents who answered "don't know" on the items contained in Table 4 ranged from 0 percent to 2 percent across all survey administrations (data not shown). The nonresponse rates in Table 4 ranged from 2 percent to 8 percent (baseline), 10 percent to 16 percent (post-test 1), and 7 percent to 9 percent (post-test 2) (data not shown).

Generally, associations between attitude/belief statements and demographic characteristics were not observed (data not shown). However, respondents in their current profession for ≥1 year were consistently more likely to perceive severe personal and public health consequences resulting from a moderate flood scenario than those having less than 1 year. At baseline, for example, levels of perceived high personal severity of the event were 11 percent for respondents with <1 year and at least 44 percent for respondents with ≥1 year in their current profession ($p = 0.046$).

Table 1. Distribution of demographic characteristics for pretraining and post-training survey administrations						
	Pretest (n = 131)		Post-test 1 (n = 129)		Post-test 2 (n = 69)	
	No.	Percent	No.	Percent	No.	Percent
Gender						
Female	84	64.1	86	66.7	49	71.0
Male	46	35.1	43	33.3	18	26.1
Nonresponse*	1	0.8	0	0.0	2	2.9
Age						
Under 29	15	11.5	15	11.6	4	5.8
30-39	22	16.8	22	17.1	12	17.4
40-49	31	23.7	31	24.0	8	11.6
50-59	35	26.7	34	26.4	27	39.1
60 or over	28	21.4	27	20.9	15	21.7
Nonresponse	0	0.0	0	0.0	3	4.3
Professional classification						
Clinical staff	21	16.0	17	13.2	12	17.4
Public health official	24	18.3	25	19.4	13	18.8
Communicable disease staff	4	3.1	5	3.9	6	8.7
Environmental health staff	13	9.9	12	9.3	8	11.6
Public information staff	8	6.1	11	8.5	5	7.2
Other public health professional staff	32	24.4	35	27.1	17	24.6
Technical support	21	16.0	16	12.4	5	7.2
Laboratory professional	21	16.0	16	12.4	1	1.4
Nonresponse	3	2.3	3	2.3	2	2.9
Current profession						
Less than 1 year	10	7.6	11	8.5	1	1.4
1-5 years	27	20.6	35	27.1	17	24.6
6-10 years	23	17.6	24	18.6	12	17.4
Greater than 10 years	61	46.6	55	42.6	37	53.6
Nonresponse	10	7.6	4	3.1	2	2.9

Table 1. Distribution of demographic characteristics for pretraining and post-training survey administrations (continued)						
	Pretest (n = 131)		Post-test 1 (n = 129)		Post-test 2 (n = 69)	
	No.	Percent	No.	Percent	No.	Percent
Current organization						
Less than 1 year	16	12.2	15	11.6	0	0.0
1-5 years	48	36.6	47	36.4	25	36.2
6-10 years	22	16.8	22	17.1	14	20.3
Greater than 10 years	42	32.1	41	31.8	29	42.0
Nonresponse	3	2.3	4	3.1	1	1.4
Highest education level						
GED	29	22.1	28	21.7	14	20.3
Bachelor's	63	48.1	59	45.7	34	49.3
Master's	26	19.8	26	20.2	13	18.8
Doctorate	6	4.6	6	4.7	3	4.3
Nonresponse	7	5.3	10	7.8	5	7.2
Directly supports family members						
Yes	77	58.8	78	60.5	37	53.6
No	54	41.2	51	39.5	31	44.9
Nonresponse	0	0.0	0	0.0	1	1.4
First responder						
Yes	70	53.4	81	62.8	41	59.4
No	48	36.6	44	34.1	24	34.8
Not sure	9	6.9	3	2.3	3	4.3
Nonresponse	4	3.1	1	0.8	1	1.4
EPPM categories†						
LT/LE	41	31.3	42	32.6	22	31.9
LT/HE	25	19.1	18	14.0	14	20.3
HT/LE	27	20.6	7	5.4	11	15.9
HT/HE	32	24.4	44	34.1	16	23.2
Nonresponse	6	4.6	18	14.0	6	8.7

*Nonresponse includes missing and "don't know" responses.
†EPPM categories: LT, low threat; HT, high threat; LE, low efficacy; and HE, high efficacy.

Table 2. Percent of respondents with "yes" responses to personal emergency preparedness questions for pretraining and post-training survey administrations

	Pretest (n = 131)		Post-test 1 (n = 129)		Post-test 2 (n = 69)	
	No.*	Percent*	No.	Percent	No.	Percent
I have a family emergency kit.	53	40.8	85	68.5	51	77.3
I feel confident that I know what should be included in a family emergency kit.	104	81.2	122	99.2	65	98.5
I feel confident that I know where to find more resources and information on what should be included in a family emergency kit.	118	91.5	125	99.2	66	100.0
I have a family emergency plan.	48	37.8	66	52.8	44	66.7
I feel confident that I know what should be included in a family emergency plan.	94	73.4	123	99.2	65	100.0
I feel confident that I know where to find more resources and information on what should be included in a family emergency plan.	117	90.0	125	99.2	65	100.0
I feel confident that I know what actions to take to protect myself and my family during different disasters that could occur in my area.	99	76.2	123	97.6	62	96.9
I feel confident that I know where to find more resources and information on the actions I need to take to protect myself and my family during different disasters that could occur in my area.	111	86.0	125	99.2	62	96.9

*Numbers and percentages do not take into account "don't know" responses or missing data.

Attitude/belief statements by EPPM category

The percent agreement for the attitude/belief statements within the EPPM categories was unchanged between the three surveys (Tables 5a, baseline; 5b, post-test 1; and 5c, post-test 2), with the exception of the perceived personal severity of the event. The percentage of respondents indicating that severe personal consequences were likely to result from a moderate flood was higher for HT respondents than LT respondents, across efficacy levels, at post-test 1 (90 percent HT vs 73 percent LT, $p = 0.030$) and post-test 2 (89 percent HT vs 19 percent LT, $p < 0.001$). The difference in percentages was smaller at baseline (58 percent HT vs 45 percent LT, $p = 0.156$).

At baseline, the percentage of respondents perceiving confidence in their personal preparedness knowledge was higher for HE respondents than LE respondents at baseline (98 percent HE vs 74 percent LE, $p < 0.001$), post-test 1 (100 percent HE vs 88 percent LE, $p = 0.006$), and post-test 2 (100 percent HE vs 85 percent, $p = 0.054$). The perceived impact of their response on their job success was also higher for HE respondents than for LE respondents at baseline (100 percent HE vs 86 percent LE, $p = 0.003$), post-test 1 (100 percent HE vs 89 percent LE, $p = 0.015$), and post-test 2 (100 percent HE vs 85 percent LE, $p = 0.054$).

The percentage of respondents who perceived an impact of their personal protective activities on the overall success of a flood response was higher in respondents categorized as HE than in respondents categorized as LE at post-test 1 (100 percent HE vs 91 percent LE, $p = 0.035$) and post-test 2 (100 percent HE vs 85 percent LE, $p = 0.054$). The difference

Table 3. Percent of participants with correct responses to knowledge questions by pretraining and post-training survey administrations						
	Pretest (n = 131)		Post-test 1 (n = 129)		Post-test 2 (n = 69)	
	No.*	Percent*	No.	Percent	No.	Percent
What is the minimum duration of food and water supply to be included in an emergency family kit?	113	89.0	120	95.2	64	95.5
In a large-scale disaster, approximately how long does it usually take for local emergency services to assist individual households.	65	51.6	110	86.6	52	78.8
In the event of a winter storm, which of the following are necessary in each household for effective response to this event?	129	100.0	127	100.0	66	98.5
When storing food for an emergency, what type of foods should be included in a family emergency kit?	123	96.1	125	99.2	63	94.0
What is the most common natural disaster in the United States?	111	86.0	126	98.4	59	89.4
Without electricity or a cold source, food stored in refrigerators and freezers can become unsafe. Bacteria in food grow at temperatures between:	88	69.8	109	87.2	41	65.1
Packaged food (enclosed in plastic wrapping) is not safe for consumption if:	120	91.6	112	88.2	56	86.2
Why should an "out-of-town" contact be identified in a family emergency plan?	90	68.7	113	88.3	50	75.8
Respondents with adequate knowledge, defined as having answered more than six questions correctly.	106	80.9	125	96.9	57	82.6
*Numbers and percentages do not take into account "don't know" responses or missing data.						

between HE and LE respondents was less pronounced at baseline (96 percent HE vs 91 percent LE, p = 0.290).

Personal preparedness statements by EPPM category

At all survey administrations, there is no evidence that HT/HE respondents were more likely than LT/LE respondents to have a kit or a communication plan or to know where to find preparedness information (Tables 6a, baseline; 6b, post-test 1; and 6c, post-test 2). At baseline, HE respondents were more informed than LE respondents about what to include in a family kit and emergency plan, what protective actions to take in a disaster, and where to find preparedness information. Within all four of the EPPM categories, the percentage of respondents reporting they had emergency kits and plans, and knew where to find preparedness information, was much higher at post-test 2 than at baseline. As the pretraining and post-training questionnaires did not have identifiers to match responses from the same subject across administrations, the EPPM category assigned to a single respondent could vary across survey administrations.

Facilitators and barriers to preparedness behaviors

Participants were asked to list the most significant factors that would i) assist and ii) prevent them from completing personal preparedness tasks *before* a

Table 4. Percent agreement with attitude and beliefs statement regarding a major flood scenario by pretraining and post-training administrations						
	Pretest (n = 131)		Post-test 1 (n = 129)		Post-test 2 (n = 69)	
	No.*	Percent*	No.	Percent	No.	Percent
A moderate flood is likely to occur in this region.†	119	92.2	111	95.7	57	89.1
If it occurs, a moderate flood in this region is likely to have severe public health consequences for affected communities.†	115	91.3	108	95.6	37	57.8
If it occurs, a moderate flood in this region is likely to have severe consequences for myself and my family.	65	50.8	91	81.2	32	50.0
I feel confident that I know what actions to take to protect myself and my family in the event of a moderate flood.	109	84.5	107	94.7	59	92.2
I feel confident that I would be able to successfully perform protective actions for myself and my family in the event of a moderate flood.‡	114	88.4	108	95.6	60	93.8
If I successfully perform protective actions for myself and my family, these actions will make a big difference to our safety during a moderate flood.‡	116	90.6	106	95.5	57	90.5
By successfully performing protective actions for myself and my family, I am also improving my ability to successfully perform my job during a moderate flood.	117	92.9	103	95.4	59	92.2
By successfully performing protective actions for myself and my family, I am contributing to the overall success of flood response in my community.	112	93.3	103	96.3	59	92.2

*Numbers and percentages do not take into account “don’t know” responses or missing data.
†Items 1 and 2 used to categorize participants as perceiving high threat or low threat according to the Extended Parallel Process Model (EPPM).
‡Items 5 and 6 used to categorize participants as perceiving high efficacy or low efficacy according to the EPPM.

moderate flood. At all survey administrations, the most commonly cited facilitating *factors* were knowledge about required tasks/access to relevant information, adequate planning and preparedness (eg, a “dry” drill with family before an incident), and cooperation from family members. Time and training were also mentioned as important facilitating factors. At all survey administrations, the most commonly cited *barriers* were lack of time, lack of money, and lack of motivation to complete preparedness tasks. Lack of storage space for supplies and lack of cooperation from family were also mentioned as important barriers to preparedness.

Course evaluation

At the end of each workshop, participants were asked to provide feedback on the training session through an evaluation form. Seventy-two participants

Table 5. Percent agreement for attitude/belief statements by EPPM category with respect to a major flood on a) pretraining administration (n = 131), b) post-training administration (n = 129), and c) 1-year follow-up administration (n = 69)								
	Low threat, low efficacy		Low threat, high efficacy		High threat, low efficacy		High threat, high efficacy	
	No.*	Percent*	No.	Percent	No.	Percent	No.	Percent
a) EPPM category (pretest)								
If it occurs, a moderate flood in this region is likely to have severe consequences for myself and my family.	20	48.8	9	37.5	16	59.3	18	56.2
I feel confident that I know what actions to take to protect myself and my family in the event of a moderate flood.	30	73.2	25	100.0	20	74.1	31	96.9
By successfully performing protective actions for myself and my family, I am also improving my ability to successfully perform my job during a moderate flood.	32	82.1	25	100.0	24	92.3	32	100.0
By successfully performing protective actions for myself and my family, I am contributing to the overall success of flood response in my community.	32	84.2	20	95.2	26	100.0	31	96.9
b) EPPM category (post-test 1)								
If it occurs, a moderate flood in this region is likely to have severe consequences for myself and my family.	34	81	10	55.6	7	100.0	39	88.6
I feel confident that I know what actions to take to protect myself and my family in the event of a moderate flood.	37	88.1	18	100.0	6	85.7	44	100.0
By successfully performing protective actions for myself and my family, I am also improving my ability to successfully perform my job during a moderate flood.	36	90	18	100.0	6	85.7	41	100.0
By successfully performing protective actions for myself and my family, I am contributing to the overall success of flood response in my community.	35	89.7	18	100.0	7	100.0	40	100.0

Table 5. Percent agreement for attitude/belief statements by EPPM category with respect to a major flood on a) pretraining administration (n = 131), b) post-training administration (n = 129), and c) 1-year follow-up administration (n = 69) (continued)

	Low threat, low efficacy		Low threat, high efficacy		High threat, low efficacy		High threat, high efficacy	
	No.*	Percent*	No.	Percent	No.	Percent	No.	Percent
c) EPPM category (post-test 2)								
If it occurs, a moderate flood in this region is likely to have severe consequences for myself and my family.	7	31.8	0	0.0	10	90.9	14	87.5
I feel confident that I know what actions to take to protect myself and my family in the event of a moderate flood.	18	81.8	14	100.0	10	90.9	16	100.0
By successfully performing protective actions for myself and my family, I am also improving my ability to successfully perform my job during a moderate flood.	17	77.3	14	100.0	11	100.0	16	100.0
By successfully performing protective actions for myself and my family, I am contributing to the overall success of flood response in my community.	18	81.8	14	100.0	10	90.9	16	100.0

*Numbers and percentages do not take into account "don't know" responses or missing data.

(55 percent) completed the evaluation. Seventy-eight percent of respondents agreed/strongly agreed that the workshop was valuable to them, and 89 percent agreed/strongly agreed that the session provided information useful immediately or in the future.

DISCUSSION

Experts generally agree that individuals will require partial or complete self-sufficiency for at least the first 72 hours during a large-scale incident.[2,23] Personal preparedness is a critical component of effective emergency preparedness, response, and recovery. Educating and empowering public-health professionals and first responders to prepare themselves and their families for emergencies may increase their WTR to emergencies and ultimately increase the preparedness level across an organization.

Workshop development

In 2007, JH-CPHP and WVBPH-CTP partnered to assess parameters of WTR and personal preparedness of West Virginia LHD employees.[11] Survey results identified gaps in employee personal preparedness which were targeted for action by WVBPH-CTP managers. These evidence-based results led to the development of a personal preparedness workshop for West Virginia LHD employees aimed at conveying important preparedness information and accomplishing key practical steps. To our knowledge, this is the first study to analyze public health workers' perceived threat and efficacy through the lens of EPPM in relationship to confidence in, and performance of, personal preparedness activities.

A strength of the syllabus is its adaptation from existing FEMA materials that are nationally standardized, comprehensive, and readily available online. Moreover, use of FEMA materials maximizes outreach of existing preparedness materials, avoids duplication, and reduces course development costs. The curriculum also complies with priorities set by the Council on Education for Public Health and Association of Schools of Public Health, such as

Table 6. Percent of respondents with "yes" responses to personal emergency preparedness questions by EPPM category on a) pretraining administration (n = 131), b) post-training administration (n = 129), and c) 1-year follow-up administration (n = 69)								
	Low threat, low efficacy		Low threat, high efficacy		High threat, low efficacy		High threat, high efficacy	
	No.*	Percent*	No.	Percent	No.	Percent	No.	Percent
a) EPPM category (pretest)								
I have a family emergency kit.	11	26.8	16	64.0	11	40.7	13	40.6
I feel confident that I know what should be included in a family emergency kit.	30	73.2	24	96.0	19	70.4	26	86.7
I feel confident that I know where to find more resources and information on what should be included in a family emergency kit.	34	83.0	24	100.0	23	85.2	32	100.0
I have a family emergency plan.	9	22.0	14	58.3	10	38.5	14	45.2
I feel confident that I know what should be included in a family emergency plan.	27	65.9	21	84.0	17	65.4	24	77.4
I feel confident that I know where to find more resources and information on what should be included in a family emergency plan.	32	78.0	25	100.0	24	88.9	31	96.9
I feel confident that I know what actions to take to protect myself and my family during different disasters that could occur in my area.	26	63.4	24	96.0	17	63.0	27	84.4
I feel confident that I know where to find more resources and information on the actions I need to take to protect myself and my family during different disasters that could occur in my area.	30	75.0	24	96.0	21	77.8	31	96.9
b) EPPM category (post-test 1)								
I have a family emergency kit.	26	65.0	11	61.1	6	85.7	32	72.7
I feel confident that I know what should be included in a family emergency kit.	42	100.0	17	100.0	6	85.7	42	100.0

Table 6. Percent of respondents with "yes" responses to personal emergency preparedness questions by EPPM category on a) pretraining administration (n = 131), b) post-training administration (n = 129), and c) 1-year follow-up administration (n = 69) (continued)								
	Low threat, low efficacy		Low threat, high efficacy		High threat, low efficacy		High threat, high efficacy	
	No.*	Percent*	No.	Percent	No.	Percent	No.	Percent
I feel confident that I know where to find more resources and information on what should be included in a family emergency kit.	42	100.0	18	100.0	6	85.7	44	100.0
I have a family emergency plan.	24	58.5	7	38.9	1	14.3	27	61.4
I feel confident that I know what should be included in a family emergency plan.	41	100.0	18	100.0	6	85.7	43	100.0
I feel confident that I know where to find more resources and information on what should be included in a family emergency plan.	42	100.0	18	100.0	6	85.7	44	100.0
I feel confident that I know what actions to take to protect myself and my family during different disasters that could occur in my area.	42	100.0	18	100.0	4	57.1	44	100.0
I feel confident that I know where to find more resources and information on the actions I need to take to protect myself and my family during different disasters that could occur in my area.	42	100.0	18	100.0	6	85.7	44	100.0
c) EPPM category (post-test 2)								
I have a family emergency kit.	18	81.8	13	92.9	6	54.6	12	75.0
I feel confident that I know what should be included in a family emergency kit.	22	100.0	14	100.0	11	100.0	16	100.0
I feel confident that I know where to find more resources and information on what should be included in a family emergency kit.	22	100.0	14	100.0	11	100.0	16	100.0
I have a family emergency plan.	15	68.2	9	64.3	7	63.6	12	75.0

Table 6. Percent of respondents with "yes" responses to personal emergency preparedness questions by EPPM category on a) pretraining administration (n = 131), b) post-training administration (n = 129), and c) 1-year follow-up administration (n = 69) (continued)								
	Low threat, low efficacy		Low threat, high efficacy		High threat, low efficacy		High threat, high efficacy	
	No.*	Percent*	No.	Percent	No.	Percent	No.	Percent
I feel confident that I know what should be included in a family emergency plan.	21	100.0	14	100.0	11	100.0	16	100.0
I feel confident that I know where to find more resources and information on what should be included in a family emergency plan.	21	100.0	14	100.0	11	100.0	16	100.0
I feel confident that I know what actions to take to protect myself and my family during different disasters that could occur in my area.	20	95.2	14	100.0	10	100.0	16	100.0
I feel confident that I know where to find more resources and information on the actions I need to take to protect myself and my family during different disasters that could occur in my area.	21	95.5	13	100.0	11	100.0	15	100.0
*Numbers and percentages do not take into account "don't know" responses or missing data.								

"creating innovative, collaborative partnerships to better utilize academic and practice resources."[24]

Workshop delivery and data interpretation

Because of the nature of the workshop, training was geographically limited to West Virginia. To allow for wider availability, after initial content delivery, all course materials were provided to WVDHHR managers for dissemination to their employees. In follow-up, WVBPH-CTP staff adapted materials to provide a "delivery toolkit" that LHD staff could modify to deliver training to other public health responders within their jurisdiction (eg, LHD staff, Medical Reserve Corps, and other health and medical volunteer groups). Two more trainings were delivered by WVDHHR to 80 members of the public. Additionally, the toolkit was used by WVDHHR in hosting a group of school children rotating through the agency as part of "take your child to work day," and proved to be a visible and interactive method of sharing information with children and their parents. The materials have also been used in agency displays and other presentations promoting volunteerism in public health and in family and community resiliency.

Given the lack of an official, nationally standardized testing or credentialing process for personal preparedness training modules, JH-CPHP conducted preparedness knowledge assessments pretraining and post-training and 1-year after the workshop. The highest average percentage of correct answers occurred at post-test 1 with lower and similar average percents at baseline and post-test 2, suggesting a surge of knowledge immediately postworkshop with no noteworthy retention of knowledge 1-year postworkshop. This finding suggests that, longitudinally, refresher courses may be beneficial. However, it should be noted that correct response rates to question 2 ("In a large-scale disaster, approximately how long does it usually take for local emergency services to assist individual households?") remained high at both post-tests 1 and 2 (Table 3). Question 2 relates to self-sustainability postdisaster and is a salient concept at the heart of personal preparedness.

Pretraining and post-training assessments gauged whether information and tools conveyed during the workshop were associated with an increase in preparedness activities or confidence postworkshop. The number of respondents reporting personal emergency kits and preparedness plans increased immediately after the workshop compared to before the workshop (post-test 1 vs baseline). The first finding may be explained by the distribution of small household preparedness items during the course of the workshop. The latter finding may be explained by an exercise in which participants discussed important components of preparedness and started drafting their plan. One-year postworkshop, the percentage of respondents who reported having personal emergency kits and preparedness plans increased from post-test 1, suggesting some respondents assembled supply kits and plans after completing the workshop (Table 2). At post-test 2, more respondents reported having kits (77.3 percent) than having emergency plans (66.7 percent). This finding is in line with several other studies reporting that people are less likely to create a family plan than assemble a supply kit.[25-28] The percentage of respondents reporting confidence in personal preparedness activities also increased from baseline to post-test 1 and was maintained or increased at post-test 2 (Table 2), suggesting that respondents felt more confident in their personal preparedness knowledge and abilities immediately after and up to 1 year following the workshop.

Of the three "get a kit, make a plan, be informed" tasks, "be informed" was the most difficult to quantify. To gain information on the "be informed" activity, two perception comments were used: "confidence in knowing what protective actions to take" and "confidence in knowing where to find more resources and information" (items 7 and 8, Table 2). These two comments are a subjective self-assessment and thus do not measure whether the activity was performed. It should also be noted that many emergency planners and academic researchers frame personal disaster preparedness around only two tasks: collecting emergency supplies and creating a preparedness plan.[2,25-28] Although these are the two most frequently observed constituents of personal preparedness activities, the researchers chose to follow the FEMA three-activity personal preparedness model in line with current US federal guidance. Furthermore, the importance of "staying informed" is well established in the professional literature, with several studies supporting the notion that people with higher threat-specific knowledge are more likely to prepare for these events.[26,29,30]

More respondents perceived severe personal consequences resulting from a moderate flood (item 3 in Table 4) at post-test 1 compared to baseline and post-test 2. This suggests that information conveyed in the workshop may have increased participants' threat perception initially; however, it was not retained 1-year postworkshop. Given there were no severe flood events in West Virginia in the year that followed the workshop, threat perception may have also been moderated by an "incident-free" year. Interestingly, respondents who were in their current profession for more than 1 year were more likely to perceive severe consequences resulting from a moderate flood than respondents in their current profession for less than a year. This may indicate increased experience working with floods or a better understanding of the potential consequences of a moderate flood. A sharp decrease in the percentage of respondents who perceived severe public health consequences resulting from a moderate flood in their region (item 2 in Table 4) was observed at post-test 2 as compared to baseline and post-test 1. This indicates a reduced threat perception by participants' 1-year postworkshop.

Personal preparedness in the context of EPPM theory

EPPM theory would suggest that those who fall into a HT/HE category would likely engage in more personal preparedness activities than in the other categories. This finding has also been shown outside the context of EPPM in studies suggesting that higher perceived risk is associated with greater rates of personal preparedness activities.[2,5,26,29,31,32] Several of our findings are consistent with the EPPM. HT respondents tended to agree more often that severe personal impacts were likely to result from a moderate flood than LT respondents at post-tests 1 and 2. HE respondents perceived confidence in their knowledge and the impact of their response on their job success at higher rates than LE respondents at all survey administrations. At post-test 1, and to a lesser degree at post-test 2, HE respondents had higher rates of perceiving an impact of their response on overall success than LE respondents. At baseline, HE respondents were more informed than LE respondents about what to include in a family emergency kit and emergency plan, protective actions to take in a disaster, and where to find information on preparedness activities. One-year postcurriculum, the level of self-reported kit-making and confidence in information-acquisition behaviors was at or near 100 percent across all EPPM profiles. The EPPM predicts that personnel fitting HT/HE profiles would consistently be more apt to engage in desired personal preparedness behaviors; however, we cannot evaluate this prediction at post-test 2 when nearly all personnel reported the same behavior. Further qualitative study, along with expanded geographical research beyond a single state, is warranted to gain additional nuanced insights.

Facilitators and barriers to preparedness behaviors

With respect to factors that could assist participants in completing household preparedness tasks, results suggest employees would benefit from more training on personal preparedness and access to easy instructions and relevant information on undertaking such tasks. Lack of time, interest and "buy-in" from family members were mentioned as both barriers and facilitators to performing preparedness behaviors. The issue of time may be simply resolved by LHD employers allocating work hours for completion of specific personal preparedness activities to support overall workforce preparedness. Family cooperation may be improved by periodic family preparedness training or "fun-days" where partners and children are encouraged to attend information sessions and preparedness-focused games—such days could also be promoted as community events. Lack of money was mentioned as an important barrier to completing preparedness tasks deemed "too expensive" such as stocking up on food and purchasing basic preparedness supplies. This barrier has important policy implications especially in West Virginia, one of the lowest income states in the United States. Given local governmental workforce cutbacks amidst fiscal austerity, and the backdrop of a growing array of public health threats, an "all-hands-on-deck" approach to preparedness is crucial. Every LHD employee should have a critical role to play in disaster response. If employees are prepared for emergencies, they are more likely to report to duty knowing their families and homes are secure. A possible solution is government funding earmarked as an investment to ensuring enhanced workforce capacity by encouraging the personal preparedness activities of LHD employees.

Limitations

Although the "Think Inside the Box Delivery Toolkit" has enabled West Virginia LHD staff to continue delivering personal preparedness trainings, workshops have generally been limited to West Virginia. Baseline, post-test 1, and post-test 2 surveys were anonymous and could not be matched to the same respondent over all survey administrations. Therefore, it was not possible to look for average individual changes over time.

The number of respondents within EPPM and demographic categories was not sufficiently large to fit regression models. Hence, we present percentages and p-values to help quantify the associations between these categories and responses to the attitudes/beliefs and personal preparedness questions. Because of the small number of responses within categories and inability to match respondents' answers across survey administrations, more extensive statistical analyses could not be conducted.

The knowledge assessment tool, composed by subject-matter experts to address core concepts of FEMA personal preparedness materials, was neither validated nor verified and test questions were not linked to specific desired competencies.

CONCLUSIONS

The ultimate goal of personal disaster preparedness is a well-trained workforce, able to serve a community that is similarly well prepared. It follows that a well-prepared household lessens (but does not necessarily fully eliminate per se) worker distraction created by concern for the safety of family members, apprehension about arriving to work safely, lack of self-efficacy, and a lack of understanding of local public health threats. In the context of the study, personal preparedness was measured through knowledge questions and use of the FEMA three-activity personal preparedness model. Overall, the workshop was well received by participants and successful in increasing engagement in personal preparedness activities, with the percentage of respondents reporting emergency kits, preparedness plans, and confidence in personal preparedness activities all increased at post-test 2 (compared to post-test 1).

Although there was a measured surge of knowledge immediately post-workshop there was no significant surge seen one-year post workshop – which may indicate there was no retention of knowledge post workshop. If so, this could identify the need for longitudinal follow-on activities to sustain and reinforce initial gains from personal preparedness trainings.

Further, the study illustrates how established behavioral models such as the EPPM can facilitate examination of, and underpin interventions to address, long-recognized gaps in personal disaster preparedness, including among responders themselves. This study not only provides initial insights but also points to still-unanswered questions about the longitudinal relationship between the EPPM's dimensions of threat and efficacy and desired personal protective behaviors among a centrally important response cadre. Given the importance of personal preparedness from a policy implementation standpoint, future research could explore these critical unanswered questions through expanded cohort analyses and mixed methods designs.

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